Columbia University

APMAE4302

Methods in computational science

Homework 3

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March 9, 2026

1 Problem 1

1) First, the f function is needed. For find it, the following exact solution is given $u_{exact}(x, y)$

$$u_{exact}(x, y) = A \exp\left(-\frac{(x - x_0)^2 + (y - y_0)^2}{\sigma^2}\right) \tag{1}$$

Setting $b = \left(-\frac{(x-x_0)^2+(y-y_0)^2}{\sigma^2}\right) \rightarrow u_{exact} = A\exp(b)$

- $\frac{\partial u}{\partial x} = A\exp(b)\frac{\partial b}{\partial x}$
- $\frac{\partial^2 u}{\partial x^2} = A\left[\exp(b)\left(\frac{\partial b}{\partial x}\right)^2 + \exp(b)\frac{\partial^2 b}{\partial x^2}\right] = A\exp(b)\left[\left(\frac{\partial b}{\partial x}\right)^2 + \frac{\partial^2 b}{\partial x^2}\right]$
- $\frac{\partial u}{\partial y} = A\exp(b)\frac{\partial b}{\partial y}$
- $\frac{\partial^2 u}{\partial y^2} = A\left[\exp(b)\left(\frac{\partial b}{\partial y}\right)^2 + \exp(b)\frac{\partial^2 b}{\partial y^2}\right] = A\exp(b)\left[\left(\frac{\partial b}{\partial y}\right)^2 + \frac{\partial^2 b}{\partial y^2}\right]$

Then, the equation is:

$$-\left(A\exp(b)\left[\left(\frac{\partial b}{\partial x}\right)^2 + \frac{\partial^2 b}{\partial x^2}\right] + A\exp(b)\left[\left(\frac{\partial b}{\partial y}\right)^2 + \frac{\partial^2 b}{\partial y^2}\right]\right) + \gamma\left(A\exp(b)\right)^p = f \tag{2}$$

$$-A\exp(b)\left(\left(\frac{\partial b}{\partial x}\right)^2 + \left(\frac{\partial b}{\partial y}\right)^2 + \frac{\partial^2 b}{\partial x^2} + \frac{\partial^2 b}{\partial y^2}\right) + \gamma\left(A\exp(b)\right)^p = f \tag{3}$$

And $\frac{\partial b}{\partial x} = -\frac{2(x-x_0)}{\sigma^2}$, $\frac{\partial^2 b}{\partial x^2} = -\frac{2}{\sigma^2}$, $\frac{\partial b}{\partial y} = -\frac{2(y-y_0)}{\sigma^2}$, $\frac{\partial^2 b}{\partial y^2} = -\frac{2}{\sigma^2}$

$$-A\exp(b)\left(\frac{4}{\sigma^4}((x - x_0)^2 + (y - y_0)^2) - \frac{4}{\sigma^2}\right) + \gamma\left(A\exp(b)\right)^p = f \tag{4}$$

$$-A\exp(b)\left(-\frac{2}{\sigma^2}b - \frac{4}{\sigma^2}\right) + \gamma\left(A\exp(b)\right)^p = f \tag{5}$$

$$\frac{2}{\sigma^2}A\exp(b)\left(b + 2\right) + \gamma\left(A\exp(b)\right)^p = f \tag{6}$$

The discretization is the following:

$$\frac{\partial^2 u}{\partial x^2} \approx \frac{u^{i-1,j} - 2u^{i,j} + u^{i+1,j}}{\Delta x^2} \tag{7}$$

$$\frac{\partial^2 u}{\partial y^2} \approx \frac{u^{i,j-1} - 2u^{i,j} + u^{i,j+1}}{\Delta y^2} \tag{8}$$

So that, the equation is:

$$-\frac{u^{i-1,j} - 2u^{i,j} + u^{i+1,j}}{\Delta x^2} - \frac{u^{i,j-1} - 2u^{i,j} + u^{i,j+1}}{\Delta y^2} + \gamma(u^{i,j})^p = f \tag{9}$$

If the grid is equally spaced, that means $\Delta x = \Delta y = h$:

$$\frac{4u^{i,j} - u^{i-1,j} - u^{i+1,j} - u^{i,j-1} - u^{i,j+1}}{h^2} + \gamma(u^{i,j})^p = f \tag{10}$$

So that the residual is:

$$F^{i,j}(u) = \frac{4u^{i,j} - u^{i-1,j} - u^{i+1,j} - u^{i,j-1} - u^{i,j+1}}{h^2} + \gamma(u^{i,j})^p - f = 0 \tag{11}$$

And for the boundary nodes:

$$F^{i,j}(u) = u^{i,j} - u_{exact}(x_i, y_i) = 0 \tag{12}$$

2) The Jacobian corresponds to the equation's derivatives with respect to the u variables. For nodes that are inside the grid:

$$\frac{\partial F^{i,j}}{\partial u^{i,j}} = \frac{4}{h^2} + \gamma p(u^{i,j})^{p-1} \tag{13}$$

$$\frac{\partial F^{i,j}}{\partial u^{i+1,j}} = \frac{\partial F^{i,j}}{\partial u^{i-1,j}} = \frac{\partial F^{i,j}}{\partial u^{i,j+1}} = \frac{\partial F^{i,j}}{\partial u^{i,j-1}} = -\frac{1}{h^2} \tag{14}$$

And for the boundary:

$$\frac{\partial F^{i,j}}{\partial u^{i,j}} = 1 \tag{15}$$

$$J(u) = 1 \tag{16}$$

If all unknowns are stacked into a global vector $\mathbf{u}$, the Jacobian for the interior equations can be written as

$$\mathbf{J}(\mathbf{u}) = \mathbf{A} + \gamma p \,\mathrm{diag}\big(\mathbf{u}^{[p-1]}\big)\,, \tag{17}$$

where $\mathbf{A}$ is the five point finite difference matrix associated with the operator $-\nabla^2$ (each row as $\begin{bmatrix} ... & -1 & ... & -1 & 4 & -1 & ... & -1 & ... \end{bmatrix}$), and $\mathrm{diag}(\mathbf{u}^{[p-1]})$ is a diagonal matrix whose entries are u_k^{p-1}.

For boundary nodes, the corresponding rows of the Jacobian are replaced by rows of the identity matrix, since the boundary residual is simply $u_k - g_k = 0$. Thus, the global Jacobian is obtained by using the interior stencil rows for interior nodes and identity rows for boundary nodes.

3)

a) For this problem, the code `reaction2d.c` was wrote and attached and uploaded to the repositori

b)The function f is controlled to be $f = -\nabla^2 u_{exact}$ when `-rct_linear_f` is set, and $f = -\nabla^2 u_{exact} + \gamma u^p$ as the following piece of code

```
PetscReal fRHS(PetscReal x, PetscReal y, AppCtx *user) {
    PetscReal f = -d2ufunction(x,y);
    if (!user->linear_f) {
        PetscReal uex = ufunction(x, y);
        f += user->gamma * PetscPowReal(uex, (PetscReal)user
           ->p);
    }
    return f;
```

c) For the residual is created from the function:

```
PetscErrorCode FormFunctionLocal(DMDALocalInfo *info,
                                 PetscReal **u,
                                 PetscReal **FF,
                                 AppCtx    *user) {
                                 .
                                 .
                                 .
                                 }
```

And the jacobian by the function:

```
PetscErrorCode FormJacobianLocal(DMDALocalInfo *info,
```

```
                                        PetscReal     **u,
                                        Mat            J,
                                        Mat            P,
                                        AppCtx        *user) {
                                        .
                                        .
                                        .
                                        }
```

d) The code can set the Jacobian option through the lines

```
    PetscCall(DMDASNESSetJacobianLocal(da,
              (DMDASNESJacobianFn *)FormJacobianLocal, &user)
                 );
    PetscCall(SNESSetFromOptions(snes));
```

The call to `SNESSetFromOptions` allows PETSc to override the analytical Jacobian registered in `DMDASNESSetJacobianLocal` with finite-difference approximations using the options -snes_fd, -snes_fd_color, -snes_mf, and -snes_mf_operator at runtime, without modifying the code.

e) Paraview option is set by the following lines:

```
    PetscCall(PetscViewerVTKOpen(PETSC_COMM_WORLD, "
       reaction2d.vtr",
                                 FILE_MODE_WRITE, &viewer));
    PetscCall(PetscObjectSetName((PetscObject)u,       "u"));
    PetscCall(PetscObjectSetName((PetscObject)uexact, "uexact
       "));
    PetscCall(VecView(u,       viewer));
    PetscCall(VecView(uexact, viewer));
    PetscCall(DMView(da,       viewer));
    PetscCall(PetscViewerDestroy(&viewer));
```

The variable u is the solution from finite difference method, and *uexact* is the exact solution. There is no RHS output because in Newton Raphson there is no RHS but residual evaluations. Note that this was extracted from `poisson2d.c` code

4) For a 65×65 grid with γ, the following result is obtained:

```
  0 SNES Function norm 7.337078385146e+02
    0 KSP Residual norm 7.337078385146e+02
    1 KSP Residual norm 6.204150044152e-11
  1 SNES Function norm 6.379557175902e-11
```

```
on 65 x 65 grid:  rel_error |u-uexact|_2/|uexact|_2 =
    0.000524515
```

The method converges in one iteration of the Newton Raphson method with a $\|F(u)\|_2 = 6.38 \times 10^{-11} < 10^{-10}$ and a relative error of 5.24×10^{-4}. The linear system at each Newton step was solved using direct LU factorization with `-ksp_type preonly -pc_type lu -pc_factor_mat_solver_type mumps`, converging in 1 KSP iteration (see `options_file_gamma0` in files at the HW3 folder)

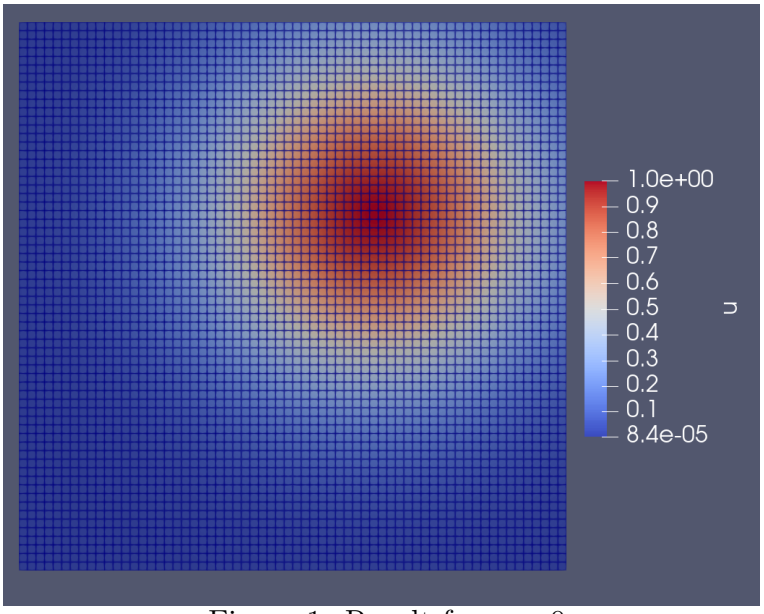

Figure 1: Result for $\gamma = 0$

5) After runing the problem with FD jacobian, these are the results:

Jacobian	Relative error	Function eval	Final Residual	Time (s)
Analytical	5.24×10^{-4}	2	6.38×10^{-11}	1.107×10^{-1}
FD (-snes_fd)	5.24×10^{-4}	4228	6.15×10^{-11}	1.430
FD Colored (-snes_fd_color)	5.24×10^{-4}	7		1.174×10^{-1}

Table 1: Performance comparison for different Jacobian strategies.

6)